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Comment on FR Doc # 2025-02305

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General Comment

The University of Virginia (UVA) welcomes the opportunity to submit a response to the Office of Science and Technology Policy Networking and Information Technology Research and Development National Coordination Office regarding the development of an Artificial Intelligence (AI) Action Plan.

To maintain U.S. leadership and dominance in global AI technological advancements, an action plan should incorporate robust support for research and workforce training activities, to be implemented in partnership with American universities. As a public research university located in Charlottesville, Virginia and founded in 1819 by Thomas Jefferson, UVA is consistently top ranked nationwide and is recognized for its rigorous academic environment, emphasis on interdisciplinary studies, and globally recognized research leadership in fields like artificial intelligence, biomedical sciences, and education. Presently, UVA has at least 275 faculty members with AI expertise. This is reflected by the university having over 160 active grants focused on AI that total more than \$166M.

The attached submission includes several recommendations for how the United States can further grow and sustain American AI dominance.

Attachments

UVA Response OSTP_NITRDNCO_031325_AZ_LM_DB

UVA Response AI RFI



Office of the Vice President
for Research

March 15, 2025

**University of Virginia Response to the Request for Information on the Development of an
Artificial Intelligence Action Plan**

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This submission includes several recommendations for how the United States can further grow and sustain American AI dominance.

Recommended Actions to Grow and Sustain U.S. AI Dominance

1. **Grow support for development and sustainment of AI-ready data and testbeds and assessment platforms.** Advances in science and technology are becoming ever more dependent on the effective deployment of safe, reliable, and secure AI systems. With the increasing numbers and types of AI systems available, AI-ready data and testbeds are essential to assess capabilities and optimize implementations and of these AI systems that the private sector is not positioned to assess themselves. Universities can serve this role since they can offer impartial, transparent, and scientifically rigorous evaluations to provide assurance that AI systems meet security and reliability standards, and that capabilities of new AI technologies or products are fully understood. This is important as it pertains to AI technologies or products developed by foreign adversaries, which may pose hidden risks, such as vulnerabilities, bias, or manipulation, which could undermine national security, privacy, and public trust. Further, as AI systems are becoming increasingly complex, universities are well placed to develop and provide comprehensive, reproducible evaluations that go beyond the scope of private sector interests. To enhance support for AI-ready data and testbeds, with the goal of ensuring that the U.S. leads in promoting secure and reliable systems, some recommended actions include:

- **Repurpose the National Institute of Standards and Technology (NIST) AI Safety Institute Consortium to leverage academic partnerships towards AI reliability and security.** This AI Safety Institute Consortium, which was established at NIST with strong bipartisan support in both the House and Senate, brings together various stakeholders, including government agencies, academic institutions, private companies, and other independent organizations, to work on creating standardized methodologies and frameworks for assessing and ensuring the safety of AI technologies. This group, in which UVA participates, could be utilized to: rapidly scale the creation of AI reliability to ensure that systems meet appropriate standards before being introduced to market; develop and promote guidelines and best practices for AI reliability and security; and to support collaborative work between sectors to address identified challenges.
- **Support AI infrastructure necessary to enable cutting edge research, including the National Artificial Intelligence Research Resource (NAIRR).** Currently in pilot form, the NAIRR is a National Science Foundation-led partnership intended to connect U.S. researchers to computational, data, software, model and training resources necessary to participate in fundamental and translational AI research. A significant focus of this work has been and should continue to be on AI trustworthiness. The NAIRR pilot, which was developed because of a task force recommendation under the first Trump Administration, is set to expire in January of 2026. The Administration should invest in the full NAIRR which will enable major increases to AI innovation by expanding access to computing power and making AI infrastructure easier to access for small companies and academic researchers.

2. **Leverage academic work for the Intelligence Community (IC) in the development and use generative AI technology to analyze open source and commercial data.** To advance U.S. national security the IC requires innovative AI tools to process the enormous amount of open-source data, with particular focus on the rapidly growing AI achievements happening worldwide in academia. AI tools have tremendous potential to produce critical insights from these open-source data useful for defense analysts and decisionmakers. For example, at UVA's [National Security Data and Policy Institute](#), researchers are working to develop tools and capabilities required to distill vast troves of information on AI technological developments in other countries available in conferences, workshops and journals. This work supports the necessary comparative analysis of technologies between nations and enables IC security analysts to acquire insights from these important open-source data and to assess possible threats to our technological leadership. The action plan should incorporate continued funding to support AI-based tools in these contexts.
3. **Leverage the potential of AI research within health and biomedical sciences.** Investing in AI research within health and biomedical sciences is essential for accelerating breakthroughs in personalized medicine, early disease detection, and treatment optimization. By leveraging AI, there is potential for the U.S. to lead the world in unlocking complex biological and health data to improve patient care, streamline healthcare systems and ultimately advance public health outcomes. The integration of AI in biomedical fields is critical not only for breakthrough treatments and diagnostics but also for improving healthcare efficiency and accessibility. Countries like China are heavily investing in AI healthcare/bioscience innovation, with initiatives such as the "Made in China 2025" plan, which emphasizes AI in medicine and biotechnology, as well as large-scale data initiatives in health. An absence of U.S. investment in this area could make the United States more reliant on China for innovations in AI-driven health and biosciences, which could create a significant security vulnerability, as critical medical technologies and sensitive health data could be controlled by a foreign power

with potentially divergent interests to ours. This could jeopardize U.S. sovereignty over its healthcare systems, compromise patient privacy, and create risks related to national security if adversarial nations influence healthcare technologies or gain access to crucial medical information. While there has been interest in AI + health related research in the United States, there are relatively few dedicated mechanisms to support work that bring together individuals with strengths in basic bioscience, public health, and clinical research with equally strong technical capabilities in computational, analytical, and systems research and the health agencies have invested comparatively less in AI-related research than physical science agencies. Some recommendations to scale-up U.S. investments in this space include:

- **Strengthen and support collaborative AI institutes between NSF and the National Institutes of Health (NIH), Centers for Disease Control and Prevention (CDC), and the Food and Drug Administration (FDA) to foster innovation and advance the integration of AI and biomedical science.** By facilitating collaboration between NSF and NIH, the United States can leverage the complementary strengths of these agencies—NSF's expertise in foundational AI research and NIH's leadership in biomedical science—to drive the development of cutting-edge AI technologies and innovations. Supporting bioscience-focused AI institutes would facilitate U.S. global leadership in AI innovation, promote cross-disciplinary research while building on the successful five-year track record of the NSF AI Institutes Program.
- **Create a Collaborative AI-Health Data Integration Research Program to develop advanced methods for the integration of AI systems with diverse healthcare data and improve health-related outcomes for all Americans.** Most American hospitals produce large amounts of data in multiple forms. These include bedside monitors that generate time stamped records, imaging, lab reports, medications, clinical notes from physicians, and electronic health records. The integration of AI systems with these different forms of health data has proved challenging because of:
 - (1) Lack of interoperability between the formats;
 - (2) Data quality issues;
 - (3) Need for privacy and security; and
 - (4) Problem complexity.

The proposed program, which could be supported by ARPA-H or the Department of Health and Human Services, could bring together interdisciplinary teams including basic and clinical researchers, computational scientists, and healthcare professionals could draw on stakeholder input from patients, policy makers, healthcare providers, hospital systems to support research: to develop AI-driven standards and interoperability solutions; to inform development of data quality and security frameworks; and to advance computational approaches for complex problem solving that can handle the complexities inherent in AI-health data integration. In addition, this research program could also support the development of curriculum, certification programs, and public-private partnerships to facilitate training of graduate students address workforce education and training needs.

4. **Harness AI-systems to improve educational instruction.** With the rapid improvements to AI technologies and tools over the past couple of years, there are significant research needs to enable the use of AI technologies to achieve their full potential to enable personalized, adaptable instruction for adults and K-12 students. Specifically, while these AI tools can enhance learning, they have not fully replaced or matched the effectiveness of skilled human teachers. Likewise, AI tools have also not been optimized as a tool to help improve instruction. Some recommendations of programs the United States could consider incorporating into

its action plan, which would help give parents and local school districts more tools to help them achieve their goals, include:

- **Support for research and development of multimodal AI-powered learning programs.** Current AI systems used in education employ text-based user interfaces. To enable interactive, adaptive, and personalized learning experiences, there is a significant need for additional research support to develop and expand multimodal AI-powered learning platforms. These multimodal AI systems will enable multiple types of inputs including text, speech, drawing/images, video, and gestures. Incorporating these multiple inputs will enhance student engagement with AI-powered learning systems and provide more dynamic, personalized, and inclusive learning environment. Such work could be supported through collaboration between the NSF and the Institute of Education Sciences.
- **AI-assisted teacher professional development.** An important approach to improving instruction uses AI systems to improve the skills and effectiveness of human instruction. This means using AI to simulate different types of students to prepare instructors for the range of learning styles and needs they will encounter in a classroom or educational environment. These AI simulations will provide a controlled environment for teachers to practice and experiment with different instructional styles to find those that can most effectively address the needs of their students. These simulations will also support improved understanding and planning for classroom dynamics and behavioral management. In addition, AI could also be utilized to develop predictive models that could identify “at risk” students.

5. **Leverage universities to prioritize development of the future AI workforce.** To maintain and strengthen the U.S.'s leadership in AI, it is critical to address the shortage of skilled professionals capable of navigating the complexities of AI integration into other sectors. Universities are well-positioned to train the next generation of experts, but they need robust federal support to enhance curricula, offer research opportunities, and facilitate professional development. By investing in these areas, the U.S. can ensure that it remains competitive in AI innovation and continues to develop solutions that improve healthcare, security, and economic outcomes. Specific recommendations include:

- **Support Graduate Fellowships and Research Programs.** Expand funding for graduate fellowships that incorporate AI, such as NSF's Research Traineeships (NRT), Graduate Research Fellowships (GRF), and CAREER awards, the Department of Energy's Computational Science Graduate Fellowship program, the NIH National Research Service Award (NRSA), and the Department of Education Graduate Assistance in Areas of National Need (GAANN) Program. These initiatives are essential for training the next generation of researchers and computational scientists, ensuring a steady pipeline of talent equipped to tackle AI challenges across sectors.
- **Promote Undergraduate Curriculum Integration.** Encourage the integration of AI, machine learning, and data science into undergraduate STEM curricula. This will provide students with a solid foundation in the mathematical principles behind AI, as well as practical experience in data science and modeling, ensuring they are well-prepared for careers in AI and related fields.
- **Foster Public-Private Partnerships for Workforce Development.** Establish partnerships between universities, industry leaders, and federal agencies to create internship programs and research opportunities that expose students to real-world AI challenges. These programs could focus on healthcare, national security, and defense-related AI applications, providing students with valuable hands-on experience in federal research environments and private industry.

- **Enhance Professional Development Opportunities.** Provide continuous education and training for professionals already in the workforce, particularly healthcare providers, data scientists, and IT specialists. Universities should be supported in offering continuing education programs that focus on emerging AI technologies, ensuring that the existing workforce remains at the forefront of AI advancements.

Sincerely,

Lori McMahon
Vice President for Research, Professor of Neuroscience



Donald E. Brown
Quantitative Foundation Distinguished Professor of Data Science

Aidong Zhang
Thomas M. Linville Professor of Computer Science



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